

Walleye pollock respond to trawling vessels

Alex De Robertis and Christopher D. Wilson

De Robertis, A., and Wilson, C. D. 2006. Walleye pollock respond to trawling vessels. — ICES Journal of Marine Science, 63: 514–522.

The potential for fish to avoid survey vessels is a major source of uncertainty in stock-assessment surveys. Although walleye pollock (*Theragra chalcogramma*) are the subject of a substantial commercial fishery in the North Pacific, their behavioural responses to approaching survey vessels remain poorly understood. As a first step in an effort to determine if walleye pollock avoid survey vessels engaged in trawling operations, we made pairwise comparisons of acoustic backscatter recorded by survey vessels while free-running and while trawling. Results are presented of acoustic backscatter recorded from NOAA's RV "Miller Freeman", which used a midwater trawl during the 1996–2002 eastern Bering Sea surveys, and a chartered commercial fishing vessel, which used a bottom trawl during a survey in 2003 in the Gulf of Alaska. In both cases, average backscatter from a vessel-mounted echosounder was significantly higher when free-running than when trawling. These decreases in backscatter are consistent with increased vessel avoidance while trawling. There were no differences in the vertical distribution of backscatter when free-running and trawling, indicating that pollock do not exhibit elevated diving responses when approached by a trawling vessel rather than by a free-running vessel. Although the study indicates that pollock respond to trawling vessels, the nature of the behavioural response cannot be determined with the methods used in this study. Future work should evaluate potential stimuli produced by trawlers to which pollock may react, and should document changes in behaviour that may occur in their presence.

Published by Elsevier Ltd on behalf of International Council for the Exploration of the Sea.

Keywords: acoustic, trawl, vessel avoidance, walleye pollock.

Received 3 March 2005; accepted 28 August 2005.

A. De Robertis and C. D. Wilson: National Marine Fisheries Service, Alaska Fisheries Science Centre, Resource Assessment and Conservation Engineering Division, 7600 Sand Point Way NE, Seattle, WA 98115, USA. Correspondence to A. De Robertis: tel: +1 206 5264789; fax: +1 206 5266723; e-mail: alex.derobertis@noaa.gov

Introduction

Fish are capable of perceiving and responding to underwater sounds generated by vessels. Fish hearing is generally most sensitive to low frequency sound in the range 10–500 Hz range, which coincides with the frequency range at which the underwater sound radiated from ships is most intense (Mitson, 1995; Popper, 2003). Field studies (e.g. Mitson, 1995; Misund, 1997; Vabø *et al.*, 2002; Mitson and Knudsen, 2003) have demonstrated that fish can respond to approaching vessels, generally by diving towards the bottom or by moving horizontally out of the vessel's path. Avoidance responses are often highly variable, with stereotyped avoidance in some situations, and no evidence of avoidance by the same species in others (CDW, unpublished data; Misund, 1997). Mounting concern about the influence of vessel avoidance during surveys has led to the construction of noise-reduced research vessels designed to minimize avoidance by the species being surveyed (Mitson

and Knudsen, 2003). Initial reports indicate that fish and krill do not avoid noise-reduced vessels (Fernandes *et al.*, 2000; Brierley *et al.*, 2003), although scientists from IMR (Norway) have recently reported that near-surface aggregations of herring can respond to a noise-reduced vessel (ICES, 2005).

Most studies on avoidance of survey vessels have focused on measuring responses to free-running (i.e. non-trawling) vessels. However, trawling vessels generally produce higher radiated noise than when free-running at the same machinery settings, so they may elicit a strong avoidance response from fish. Although published measurements of the radiated noise of operating vessels engaged in trawling are scarce, increases in radiated noise of 5–15 dB when trawling have been reported (reviewed in Mitson, 1995). Often this is the result of increased propeller and engine noise, but there is also potential for increased noise from hydraulic machinery and vibration of the trawl warps and gear. Trawling is an integral part of many fisheries

surveys (Gunderson, 1993), so given the potential for increased radiated noise while trawling, it is likely that vessel avoidance may be increased during trawling.

Fish have been documented to respond to a trawling vessel well before they come into contact with the trawl gear. Ona and Godø (1990) used a stationary echosounder to measure reactions to an oncoming trawling vessel and compared acoustic records made during bottom trawling with adjacent records made while the vessel was free-running. They found that fish performed a diving response and avoided the trawler well before encountering the trawl in water depths of up to 200 m. Target tracking of individual fish using a buoy-mounted echosounder has led to similar conclusions: fish assemblages dominated by Atlantic cod (*Gadus morhua*) respond to an approaching trawling vessel at distances of ~400 m by diving towards the bottom (Handegard *et al.*, 2003). The same authors also reported a second phase of avoidance behaviour: shortly after the vessel passed over the fish, an increase in horizontal swimming was observed coincident with the passage of the warps. This was interpreted as a two-part avoidance response to the trawler: an initial diving response to the vessel's radiated noise prior to the vessel's arrival, followed by horizontal herding attributable to visual reaction to the trawl warps and gear.

Vessel avoidance will alter the availability of fish to different techniques such as acoustic or trawl surveys. Fish that swim out of the way of an approaching vessel will not be detected by an echosounder, and fish may swim away from an oncoming trawler before they interact with the trawl gear itself. Changes in vertical distribution resulting from vessel avoidance can markedly alter the probability of detection by echosounders, as well as availability to the trawl gear (Ona and Mitson, 1996; Aglen *et al.*, 1999). Changes in tilt angle of fish responding to a trawling vessel will result in substantial changes in target strength, which bias acoustic-survey results if unaccounted for (Barange and Hampton, 1994; Misund, 1997). If responses to approaching vessels are age- and size-dependent, this will, in turn, bias estimates of size and age composition of a population. Similarly, when responses are species-specific, the relative species composition of assemblages sampled with the same gear will be biased.

In this study, we analyse echosounder records to investigate if walleye pollock, *Theragra chalcogramma*, respond to trawling survey vessels. The species is a dominant groundfish in the subarctic Pacific Ocean and is the target of one of the world's largest fisheries, with 1998–2002 landings accounting for 3.8–6.2% of the annual global catch of marine fish (Bailey *et al.*, 1999; FAO, 2004). Acoustic and bottom-trawl surveys in conjunction with information from the fishery are used to assess the stock. It is currently unknown whether pollock respond to vessels engaged in trawling. As described above, if they do respond to trawling survey vessels, there is potential to influence stock-assessment survey results. As a first step in an effort to determine

whether walleye pollock avoid survey vessels engaged in trawling, we made pairwise comparisons of acoustic backscatter recorded by survey vessels while free-running and while trawling.

Methods

To assess the potential for walleye pollock to avoid trawling vessels acoustic backscatter recorded when a vessel was free-running and when it was engaged in trawling, were compared. Our approach is modelled after the method of Ona and Godø (1990), who examined echo integration data recorded while bottom trawling and compared them with records made while the same vessel was free-running. The fundamental assumption underlying this method is that if there is no increase in avoidance when trawling, the backscatter observed when free-running in the vicinity of a trawl station will be statistically indistinguishable from that recorded while trawling. In the present study, we made comparisons for two vessels representative of the vessels used by the US National Marine Fisheries Service (NMFS) to survey pollock in the North Pacific: a research vessel towing a pelagic trawl, and a chartered commercial vessel towing a bottom trawl. Paired comparisons of acoustic backscatter recorded by vessel-mounted echosounders were made when the vessels were free-running and trawling in the same area. All analyses of acoustic data were conducted using Echoview 3.10 software.

The first data set to be analysed was collected on NOAA's RV "Miller Freeman". This 66-m stern trawler is equipped with a 1800-hp main engine and is used extensively for acoustic surveys of walleye pollock in the North Pacific. Its radiated-noise signature substantially exceeds the ICES radiated-noise guidelines for noise-reduced research vessels. Towing a trawl increases underwater radiated noise, compared with the same mechanical conditions when free-running (Figure 1). Measured noise levels when free-running at 11.2 knots, a typical acoustic-survey speed, and trawling at 3.3 knots are similar. Although the avoidance reactions and hearing capability of pollock remain poorly understood, calculations based on the method of Mitson (1995) and assuming that pollock have similar hearing capabilities to Atlantic cod, which are in the same family, suggest that they could react to the vessel at ranges of ~200 m when trawling and free-running. Note that radiated noise while trawling was only measured in calm weather at engine and propeller-pitch settings that minimize the noise levels recorded on vessel-mounted hydrophones, which is not the case during routine trawling. For vessels such as the "Miller Freeman", which have a controllable-pitch propeller, radiated noise can change dramatically with small propeller-pitch adjustments (Mitson and Knudsen, 2003), and the radiated-noise levels under field conditions can vary substantially. Therefore, it is likely that when the vessel is trawling under typical conditions, radiated noise is elevated in at least some of the low frequency bands in which fish

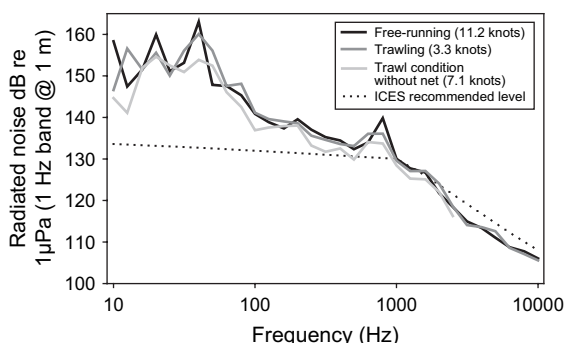


Figure 1. The underwater, radiated-noise signature of the RV “Miller Freeman” while free-running at its survey speed of 11.2 knots, while towing a pelagic trawl at 3.3 knots, and while operating under the same machinery conditions as when trawling, but not towing a net. The trawling condition represents the quietest machinery settings determined using onboard hydrophones, and may underestimate noise during typical trawling conditions. The ICES recommendation for research-vessel radiated noise (Mitson, 1995) is given for reference. Noise measurements were conducted at an US Navy acoustic range in Behm Canal, Alaska. Measurements have been converted from 1/3 octave band to 1-Hz band measurements under the assumption that the power distribution was uniform within each band. It is possible that tones within a band will exceed that shown.

hearing is most sensitive compared with when free-running at acoustic-survey speeds.

Acoustic records collected during trawl sampling on five eastern Bering Sea pollock acoustic surveys conducted by the Alaska Fisheries Science Centre from 1996 to 2002 (e.g. Honkalehto *et al.*, 2002) were compared with acoustic records collected prior to trawling in the same general area. Acoustic data were collected from 0.5 m off bottom to 14 m from the surface with a calibrated 38 kHz SIMRAD EK500 echosounder. The primary trawl used to sample pollock was an Aleutian wing pelagic trawl (AWT) with a headrope of 81.7 m. We selected cases for further analysis where AWT catches were >90% pollock by weight, and when the ship passed within 1 km of the centre of the towing path within 2 h prior to trawling. Acoustic backscatter recorded during the trawls was compared with a segment of acoustic track-line collected while the vessel was free-running. The free-running segment was selected such that it was of equal length to the distance fished by the trawl ($\bar{X} \pm \text{s.d.}$, 1.7 ± 1.2 km) and closest in space to the trawl path within a 2-h window prior to the trawl haul. Cases in which mean bottom depth differed by >1 m between the trawling and free-running segments were excluded from the analysis. Over the five surveys, 180 trawl stations broadly distributed in the eastern Bering Sea met these criteria and were

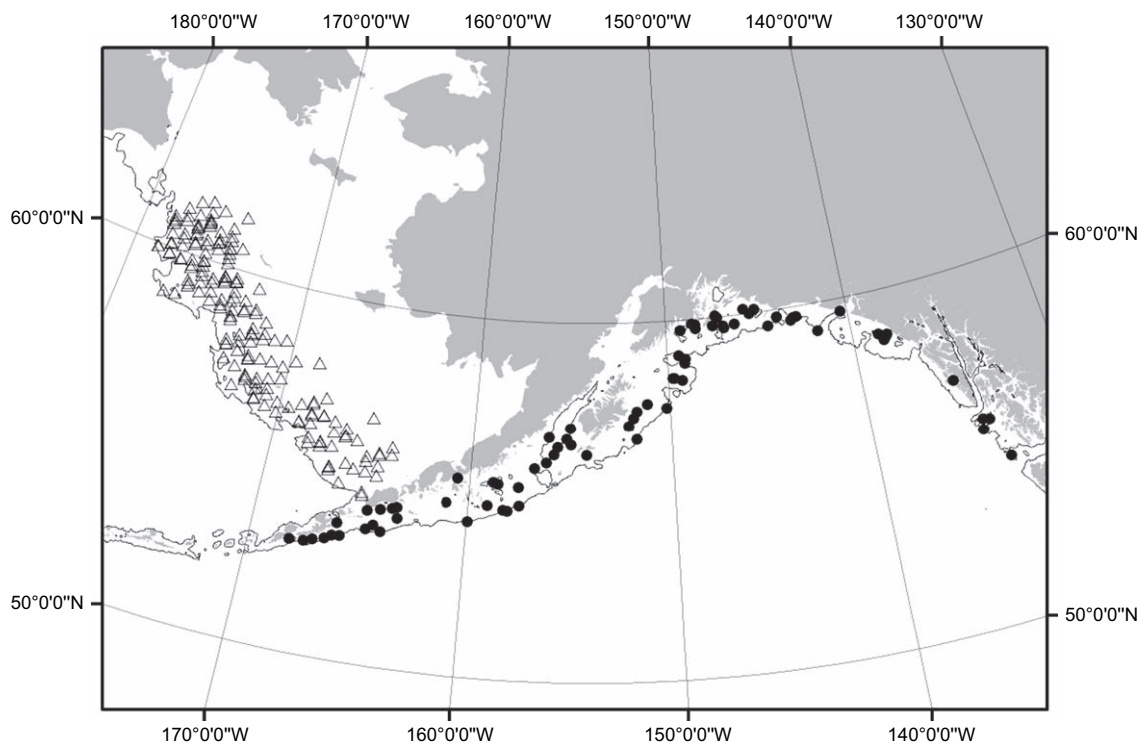


Figure 2. The locations at which comparisons of acoustic backscatter while free-running and trawling were made. Triangles indicate 1996–2002 RV “Miller Freeman” trawl locations in the eastern Bering Sea, and circles indicate stations occupied by FV “Sea Storm” in the Gulf of Alaska in 2003. The 200-m isobath is indicated by the thin black line.

included in the analysis (Figure 2). Most (156) were made in daylight. Preliminary analysis indicated that the results of statistical tests were similar if the 24 night-time trawls were excluded, and further analysis was based on pooled day and night records. Bottom depths at trawl stations averaged 116 m and ranged between 68.5 and 167.8 m. Vessel speed averaged ($\pm$ s.d.) 3.4 ± 0.7 knots while trawling, and 11.8 ± 0.9 knots while free-running. Sea conditions were generally good. Wind speeds, which are indicative of sea conditions, are available for tows conducted in 2000 and 2002, and averaged ($\pm$ s.d.) 12.3 ± 6.0 knots, with 5% of the trawls conducted in winds >20 knots. Pollock were primarily sub-adults and adults, mean length in the trawl catch averaging ($\pm$ s.d.) 36.7 ± 8.2 cm. We assigned the echosign to different categories on the basis of echosign appearance and the catch of the targeted trawl: walleye pollock, which were mostly near the bottom, and a near-surface fish/plankton mix. A third category, water-column backscatter, was used to describe backscatter attributed to walleye pollock and the fish/plankton mix combined.

We conducted a similar analysis on acoustic data collected on the commercial fishing vessel “Sea Storm” during the NMFS 2003 Gulf of Alaska bottom-trawl survey. The vessel is a 38-m stern trawler powered by a 1710-hp main engine and with a controllable-pitch propeller, and is typical of vessels chartered to conduct bottom-trawl surveys in Alaska. As with most commercial vessels, the underwater radiated-noise signature of this vessel is unknown. During this survey, the vessel used a four-panel Poly-Nor’Eastern bottom trawl with a headrope of 27.2 m. Trawls were conducted at pre-determined stations based on a stratified-random survey design (Britt and Martin, 2001). The vessel was equipped with a SIMRAD ES60 38 kHz echosounder, which was used to collect acoustic backscatter from 0.5 m off bottom to 14 m from the surface at the maximum ping rate. The echosounder was not calibrated, so nominal settings supplied by the manufacturer were used in data processing.

The acoustic data from the “Sea Storm” were used in a relative sense for comparisons of backscatter while the vessel was trawling and free-running in the same area. Sections of trackline for comparison were determined in the same manner as described earlier. As in the “Miller Freeman” analyses, the free-running trackline selected for comparison was of the same length as the distance trawled ($\bar{X} \pm$ s.d., 1.5 ± 0.6 km), and was constrained to be within 2 h and 1 km of the trawl location. Because the bottom trawl is less effective at capturing semi-pelagic pollock, and bottom depth in this area is more variable, we included trawls where pollock were captured, and made trawl and free-running comparisons in which mean bottom depth varied by $<10\%$ of bottom depth. Given these criteria, 74 trawl stations broadly distributed over the Gulf of Alaska were included in the analysis (Figure 2). All trawl hauls were conducted during daylight, and bottom depths averaged 154.9 m (range 39.5–404.2 m), with 92% of hauls

shallower than 250 m. Estimated wind speeds for trawls included in the study averaged ($\pm$ s.d.) 12.2 ± 7.8 knots, with 9% of trawls conducted in winds >20 knots. Vessel speed averaged ($\pm$ s.d.) 3.0 ± 0.1 knots while trawling and 8.1 ± 1.7 knots while free-running. Pollock were primarily sub-adults and adults, with mean length in the catch averaging ($\pm$ s.d.) 38.3 ± 12.3 cm. Because of reduced certainty in the identity of acoustic scattering on this vessel compared with the “Miller Freeman” data, backscatter was divided into a demersal category <25 m off bottom, and a pelagic category >25 m off bottom, rather than species-specific backscatter. The demersal-backscatter category is more likely to represent pollock echosign than the pelagic echosign: in a concurrent 2003 echo integration trawl survey of the Gulf of Alaska shelf ($\sim 160^\circ\text{W}$ to 147°W), pollock were estimated to account for 44% of acoustic backscatter within 25 m of the bottom, and 14% of backscatter from 25 m off bottom to the surface (M. Guttormsen, NOAA, AFSC, pers. comm.). An additional backscatter category, combined water-column backscatter, was used to describe all backscatter, and is equivalent to the demersal layer added to the pelagic layer.

The ES60 echosounder used on the “Sea Storm” is known to have a periodic systematic error that in the worst case scenario can result in 1 dB (23%) difference in acoustic measurements made on single pings, although the errors in echo-integration measurements will be substantially less as increasing numbers of pings are averaged (T. Ryan and R. Kloser, CSIRO, pers. comm.). In the context of this study, this error adds random variability to the measurements made while free-running and trawling. We conducted a numerical simulation to evaluate the effect of this error on our study, and found that given the number of pings averaged, the systematic error would introduce average uncertainty of 0.2 and 2.2% into the acoustic measurements made while the “Sea Storm” was trawling and free-running, respectively. Given such a relatively small effect, no correction for this error was made.

Paired comparisons between backscatter observed while free-running and trawling were conducted in the same manner for both data sets. To facilitate comparison of trawl stations spanning a wide range of fish density, the nautical area scattering coefficient (s_A), which is a standard measure of integrated acoustic backscatter (see MacLennan *et al.*, 2002) observed while free-running and trawling was scaled to the mean value observed at a given trawl station:

$$s_{A_scaled} = s_{A_obs} / (0.5(s_{A_free-run} + s_{A_trawl})), \quad (1)$$

where s_{A_obs} is the observation in question (either free-running or trawling s_A), $s_{A_free-run}$ is the s_A observed while free-running at that station, and s_{A_trawl} is the s_A observed while trawling at that station. The parameter s_{A_scaled} varies over the range 0–2, with 1 indicating equal backscatter while free-running and trawling. The value of s_{A_scaled} was computed for the free-running and trawling sections

of trackline selected at each trawl station. A Wilcoxon Signed-Ranks test was used to test for differences in paired s_A measurements made while trawling and free-running. This non-parametric test is based on paired rank-order statistics (Zar, 1999), and thus gives identical results for the original and the scaled observations. Comparisons were made separately for each category of backscatter identified above.

We used the “height off bottom” of acoustic backscatter as an index of the vertical distribution of fish during trawling and free-running operations. To compute this index, we averaged s_A into 1-m vertical layers, and calculated the cumulative sum of s_A from the bottom. The “height off bottom” at which 50% of the total cumulative s_A in each record is reached was used as an index of the vertical distribution of the acoustic backscatter in each record. A Wilcoxon Signed-Ranks test was used to test for paired differences in height off bottom in the trawling and free-running records. Comparisons were made separately for each category of backscatter.

Results

Paired comparison of acoustic backscatter collected while the “Miller Freeman” was free-running and trawling (Figure 3) indicated that significantly lower (16% on average) s_{A_scaled} was attributed to pollock while trawling than while free-running ($p < 0.005$). No difference in scaled backscatter was observed for the near-surface fish-plankton mix ($p > 0.05$). Water column s_{A_scaled} was significantly

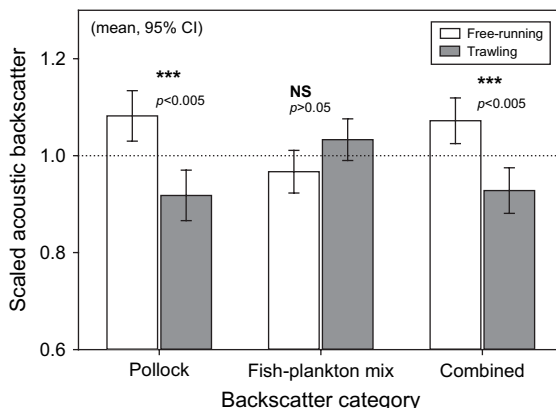


Figure 3. Paired comparisons of acoustic scattering observed while the RV “Miller Freeman” was towing a midwater trawl and free-running in the same area. To allow comparison of results over a wide range of fish densities, acoustic backscatter has been scaled to the mean backscatter observed in each of the locations compared ($n = 180$). Acoustic backscatter has been scrutinized into three categories: pollock, a near-surface fish/plankton mix, and backscatter attributed to pollock and the fish/plankton mix combined. The horizontal line indicates the null hypothesis of no difference in backscatter. The results of Wilcoxon Signed-Ranks tests for differences in backscatter while free-running and trawling are indicated.

($p < 0.005$) lower (14% on average) than trawling s_{A_scaled} . The decrease in water-column backscatter was driven largely by changes in pollock backscatter. No differences in vertical distribution of backscatter in the paired trawling vs. free-running comparisons were identified for pollock, near-surface fish/plankton mix, or water-column backscatter categories (Figure 4; $p > 0.05$ in all cases).

To explore factors that may have contributed to the discrepancy in backscatter when trawling vs. free-running, we tested for correlations between this discrepancy and factors that have been hypothesized to influence avoidance response (Table 1). There was no correlation between pollock s_{A_scaled} while free-running and water depth, pollock “height off bottom”, pollock backscatter, or mean pollock length (Table 1). This indicates that, over the ranges observed in this study, water depth, pollock vertical distribution, pollock density, and pollock size did not exert a strong effect that could be related directly to the observed discrepancy between pollock backscatter recorded while free-running and trawling.

Paired comparison of acoustic backscatter observed when the commercial vessel “Sea Storm” was free-running vs. when it was trawling identified consistent differences in backscatter during the two operations (Figure 5). Significantly lower (22% on average) s_{A_scaled} was observed in the < 25 m off-bottom demersal layer while trawling than when free-running ($p < 0.05$). Scaled backscatter in the pelagic stratum (> 25 m off bottom) was also significantly lower (15% on average) when trawling than when free-running ($p < 0.005$). Not surprisingly, the combined water column s_{A_scaled} was significantly lower (22% less) when

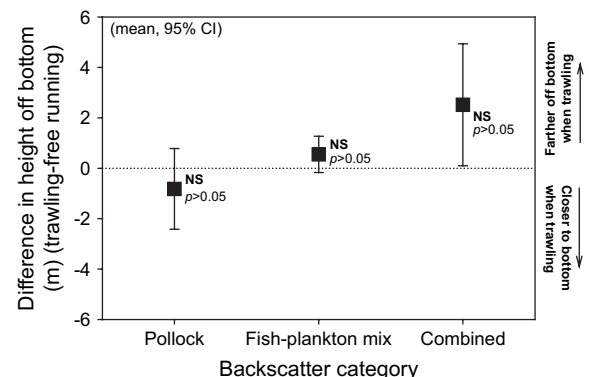


Figure 4. The difference in “height off bottom” at which 50% of the cumulative water-column backscatter was observed when the RV “Miller Freeman” towed a midwater trawl compared with paired observations while free-running in the same area. The results are presented as “height when trawling” minus “height when free-running”. Acoustic backscatter has been scrutinized in three categories: backscatter attributable to pollock, a near-surface fish/plankton mix, and backscatter attributed to pollock and the fish/plankton mix combined. The results of Wilcoxon Signed-Ranks tests of differences in “height off bottom” are indicated.

Table 1. Correlations of environmental and biological features with free-running s_{A_scaled} attributed to walleye pollock recorded by the RV “Miller Freeman”. Spearman-rank correlation coefficients (r_s) are reported for water depth, “height off bottom” at which 50% of the cumulative pollock backscatter was observed, s_A attributed to scattering from pollock, and mean pollock length in trawl catches (ns = not significant). Water depth, “height off bottom”, and s_A used in these correlations are averages of paired free-running and trawling segments. A significant correlation indicates that the discrepancy between acoustic records collected while free-running and trawling is related to this variable. If these correlations were to be repeated using trawling s_{A_scaled} , one would arrive at the same correlation coefficient with an opposite sign.

Variable	Range of observations (mean)	Correlation with free-running s_{A_scaled} (r_s)	p
Depth (m)	68.3–167.6 (116.3)	−0.045	ns
Pollock height off bottom (m)	2.6–137.0 (23.9)	0.106	ns
s_A (m ² nautical mile ^{−2})	91.3–25 177.8 (2303.6)	0.01	ns
Pollock length (cm)	16.0–52.2 (36.7)	−0.044	ns

trawling than when free-running ($p < 0.005$). No differences in vertical distribution among any of the backscatter categories were identified (Figure 6; $p > 0.05$ in all cases) between free-running and trawling records.

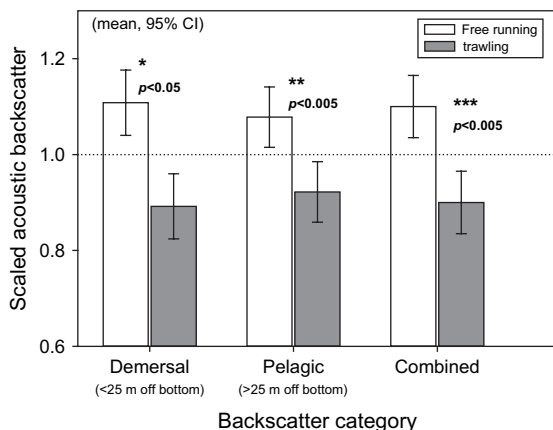


Figure 5. Paired comparisons of acoustic scattering observed while the fishing vessel “Sea Storm” is towing a bottom trawl and free-running in the same area. To allow comparison of results over a wide range of fish densities, acoustic backscatter has been scaled to the mean backscatter observed in each of the locations compared ($n = 74$). Acoustic backscatter has been separated into three layers: a demersal layer within 25 m of the bottom, a pelagic layer >25 m above bottom, and a combined layer encompassing the entire water column. The horizontal line indicates the expectation of no difference in backscatter. The results of Wilcoxon Signed-Ranks tests of differences in backscatter while free-running and trawling are indicated.

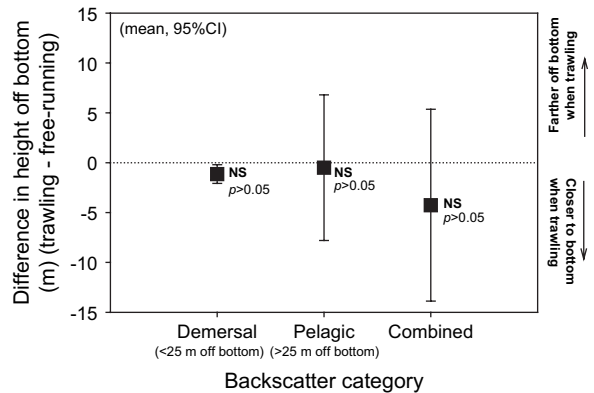


Figure 6. Difference in “height off bottom” at which 50% of cumulative water column backscatter was observed when the FV “Sea Storm” towed a bottom trawl compared with observations while free-running in the same area. Results are presented as height when trawling minus height when free-running. Acoustic backscatter has been scrutinized in three strata: demersal backscatter within 25 m of the bottom, pelagic backscatter >25 m off bottom, and backscatter in the entire water column. The results of Wilcoxon Signed-Ranks tests of differences in “height off bottom” are indicated.

We tested for correlations between s_{A_scaled} in the demersal layer and environmental and biological variables (Table 2). There was no correlation between s_{A_scaled} observed when free-running and water depth, “height off bottom” in the demersal layer, and the intensity of backscatter in

Table 2. Correlations of environmental and biological features with free-running s_{A_scaled} observed in the <25 m off-bottom demersal layer recorded by FV “Sea Storm”. Spearman-rank correlations are reported for water depth, “height off bottom” at which 50% of the cumulative pollock backscatter was observed, relative backscatter observed in the demersal layer, and mean pollock length in trawl catches (ns = not significant). Water depth, “height off bottom”, and relative backscatter used in these correlations are averages of paired free-running and trawling segments. A significant correlation indicates that the discrepancy between acoustic records collected while free-running and trawling is related to this variable. If these correlations were to be repeated using trawling s_{A_scaled} , one would arrive at the same correlation coefficient with an opposite sign.

Variable	Range of observations (mean)	Correlation with free-running s_{A_scaled} (r_s)	p
Depth (m)	39.2–399.9 (154.1)	−0.089	ns
Demersal scatter “height off bottom” (m)	5.2–21.6 (12.3)	0.024	ns
s_A (uncalibrated)	5.9–625.62 (71.7)	0.040	ns
Pollock length (cm)	17.0–58.5 (38.3)	−0.248	<0.05

the demersal layer (Table 2). However, we observed a weak, but significant, negative correlation between the mean length of pollock in trawl catches and s_{A_scaled} recorded while free-running. This indicates that, overall, there was a tendency towards a larger discrepancy between free-running and trawling backscatter when smaller pollock were present.

Discussion

Paired comparisons of acoustic backscatter from two vessels over a broad area over several years indicate a consistent reduction in scattering attributed to walleye pollock while trawling under survey conditions. Results for a research vessel towing a pelagic trawl and a commercial vessel towing a bottom trawl were similar: when trawling, scaled backscatter directly below the vessel was reduced by ~15–20% compared with when the vessel was free-running. This suggests that the fish respond to the vessel–trawl combination by the time the vessel passes over the fish aggregations, which is well before trawl passage. The disturbed fish may either move out of the path of the vessel, or change their orientation and tilt angle, which would result in reduced acoustic backscatter. Given that in the “Miller Freeman” data there was a consistent reduction in pollock backscatter, but not in the near-surface (generally <40 m from the surface) fish/plankton category, this points strongly towards a change in behaviour of pollock rather than a change in acoustic conditions or echosounder performance while trawling. In the case of the “Sea Storm” analyses, where the identity of acoustic scatterers is less certain, a consistent reduction of backscatter was observed for both demersal backscatter (within 25 m of the bottom) and pelagic (>25 m off bottom) backscatter. The observed decrease in acoustic backscatter observed during trawling is consistent with avoidance behaviour elicited by increased underwater radiated noise during trawling. However, we did not observe a consistent change in depth distribution of backscatter while trawling, which would be consistent with increased diving behaviour in response to the approaching trawling vessel.

Correlation analyses did not identify strong relationships between the discrepancy in free-running and trawling backscatter and biological or environmental characteristics. The observed decrease in average backscatter while trawling was uncorrelated with water depth, fish vertical distribution, or fish abundance. For the “Miller Freeman” data, s_{A_scaled} was unrelated to fish length, whereas for the “Sea Storm” data, there was an indication that the discrepancy between trawling and free-running backscatter may be larger for smaller fish. This supports recent observations of pollock response to a free-running vessel, which suggest that smaller fish may be more likely to exhibit vessel avoidance (CDW, unpublished). These correlations should be taken as preliminary, because the paired analyses incorporate both spatial and temporal variability, and encompass

a limited range of conditions, which may mask relationships between these factors and vessel avoidance.

It is important to recognize that the observations reported in this study are based on paired comparisons, which should only be interpreted in a relative sense. Differences identified in the comparisons will only serve to indicate if trawling increases vessel avoidance compared with free-running, but cannot reveal the extent of avoidance in comparison to the undisturbed state. For example, one cannot conclude that pollock do not dive in response to an approaching vessel, as has been reported for other species (e.g. Vabø *et al.*, 2002; Handegard *et al.*, 2003), but rather only that there is no change in depth distribution at the time of vessel passage when trawling compared with when free-running.

The targeted trawling conducted during acoustic surveys may influence the results from the RV “Miller Freeman”. The decision to trawl on such surveys is generally made on an *ad hoc* basis by examination of features observed in the echogram, and these are often associated with regions of locally elevated acoustic backscatter. If the decision to trawl is consistently made when elevated backscatter is observed while free-running, this may bias the sections of free-running trackline selected for comparison with trawling segments that have elevated backscatter compared with backscatter at the trawl locations. Although it is difficult to address this concern and to evaluate its influence, analyses of spatial and temporal variance of pollock in repeat acoustic transects in the eastern Bering Sea indicate that pollock densities do not change appreciably over temporal scales of 4 h, and that pollock aggregations are of consistent density at spatial scales of 2.5 km (Horne and Walline, 2005). Our free-running segments of trackline are within 1 km and 2 h of trawls, and effects attributable to temporal and spatial mismatches should be relatively minor. Biases introduced by targeted sampling, if they occur, are a concern for the interpretation of results from the “Miller Freeman”. However, this should not be a concern for the commercial fishing vessel because the “Sea Storm” conducted trawls at pre-determined locations according to a random stratified design.

The cues initiating avoidance from oncoming vessels are thought to be primarily auditory (Mitson and Knudsen, 2003), but other cues such as changes in light intensity associated with the presence of a vessel also have the potential to trigger avoidance responses (Misund, 1997). Although fish have highly developed vision (Douglas and Hawryshyn, 1990), it is unlikely that visual cues can explain our observations of reduced backscatter when trawling. Fish exhibit stereotyped behavioural changes when they visually detect approaching trawl gear (see review in Wardle, 1983, 1993). However, underwater visibility is limited to ranges of the order of tens of metres even in clear water owing to the rapid degradation of visual contrast (Douglas and Hawryshyn, 1990; Wardle, 1993), and it is unlikely that visual cues from a trawl trailing substantially further behind the vessel than the fish can be expected to be able to see could account for our observations. Visual

responses to trawl gear are by no means unimportant, but they are likely to dominate fish behaviour after the ship passes over the fish and the trawl gear comes into close proximity with the fish (Wardle, 1993; Handegard *et al.*, 2003).

Fish behaviour remains one of the largest potential sources of bias in fisheries surveys (Aglen, 1994; Misund, 1997). If, as suggested by our comparisons, walleye pollock consistently respond to vessels towing pelagic and bottom trawls, such behaviour has the potential to influence survey estimates of population biomass. At this point it is unclear if the decrease in acoustic backscatter while trawling is due to horizontal avoidance or changes in backscatter from individual fish. Although the paired comparisons used in this study provide statistical evidence in support of the hypothesis that trawling results in increased vessel avoidance, the analysis cannot be used to determine the stimuli or behavioural mechanisms involved in vessel avoidance. One possibility is that pollock change their tilt distribution in response to an oncoming trawling vessel more than they would to a free-running vessel. This would result in lower target strengths when trawling, as observed for Cape horse mackerel off southern Africa (Barange and Hampton, 1994), but it may not substantially influence the number and size distribution of fish sampled by a trawl. The relative contribution of changes in target strength and horizontal avoidance to our observations remains an open question, because analysis of *in situ* target strengths is not possible in this study; pollock were typically >75–100 m away from the transducer in dense near-bottom aggregations, which makes it difficult to make valid target-strength measurements of this species during the day (Traynor, 1996).

In order to definitively identify the mechanisms controlling pollock responses to approaching vessels, more detailed behavioural measures are required. To this end, we have designed and constructed an acoustic buoy similar to that described by Handegard *et al.* (2003), which we plan to use to study pollock responses to passing trawling vessels. With this type of instrument, changes in volume backscattering can be correlated with various stages of vessel passage (Ona and Godø, 1990; Vabø *et al.*, 2002), and the behaviour of individual fish can be inferred by tracking individual acoustic targets and measuring *in situ* target strength (Handegard *et al.*, 2003). In addition, further progress in understanding the mechanisms triggering avoidance can be made by measuring the radiated-noise field of trawling vessels, and integrating these measurements with knowledge of the sensory capabilities of pollock.

Acknowledgements

We are indebted to L. N. Andersen of SIMRAD, Inc., for tutelage in how to handle large data files, and to D. Somerton for providing access to the acoustic data from the Gulf of Alaska trawl survey. K. Williams assisted with preparation of Figure 2, and in analysis of trawl catches. The

comments of D. Somerton, M. Dorn, N. Williamson, and P. Von Szalay improved the manuscript. We also express our appreciation to the scientists and crews aboard NOAA's RV "Miller Freeman" and FV "Sea Storm" for their efforts at sea.

References

- Aglen, A. 1994. Sources of error in acoustic estimation of fish abundance. *In* Marine Fish Behaviour in Capture and Abundance Estimation, pp. 107–133. Ed. by A. Ferno, and S. Olsen. Fishing News Books, Oxford.
- Aglen, A., Engas, A., Huse, I., Michalsen, K., and Stensholt, B. K. 1999. How vertical fish distribution may affect survey results. *ICES Journal of Marine Science*, 56: 345–360.
- Bailey, K. M., Quinn, T. J., Bentzen, P., and Grant, W. M. 1999. Population structure and dynamics of walleye pollock, *Theragra chalcogramma*. *Advances in Marine Biology*, 37: 179–255.
- Barange, M., and Hampton, I. 1994. Influence of trawling on *in situ* estimates of Cape horse mackerel (*Trachurus trachurus capensis*) target strength. *ICES Journal of Marine Science*, 51: 121–126.
- Brierley, A. S., Fernandes, P. G., Brandon, M. A., Armstrong, F., Millard, N. W., McPhail, S. D., Stevenson, P., Pebody, M., Perret, J., Squires, M., Bone, D. G., and Griffiths, G. 2003. An investigation of avoidance by Antarctic krill of RRS James Clark Ross using the Autosub-2 autonomous underwater vehicle. *Fisheries Research*, 60: 569–576.
- Britt, L. L., and Martin, M. H. 2001. Data Report: 1999 Gulf of Alaska bottom-trawl survey. U.S. Department of Commerce, NOAA Technical Memorandum, NMFS-AFSC-121. 249 pp.
- Douglas, R. H., and Hawryshyn, C. W. 1990. Behavioural studies of fish vision: an analysis of visual capacities. *In* The Visual System of Fish, pp. 373–418. Ed. by R. Douglas, and M. B. A. Djamgoz. Chapman & Hall, London. 414 pp.
- Fernandes, P. G., Brierley, A. S., Simmonds, E. J., Millard, N. W., McPhail, S. D., Armstrong, F., Stevenson, P., and Squires, M. 2000. Fish do not avoid survey vessels. *Nature*, 404: 35–36.
- FAO. 2004. FAO Capture Production. Yearbook of Fishery Statistics. 94/1. FAO, Rome. 654 pp.
- Gunderson, D. R. 1993. Surveys of Fisheries Resources. John Wiley, New York. 248 pp.
- Handegard, N. O., Michalsen, K., and Tjøstheim, D. 2003. Avoidance behaviour in cod (*Gadus morhua*) to a bottom-trawling vessel. *Aquatic Living Resources*, 16: 265–270.
- Honkalehto, T., Williamson, N. J., de Blois, S., and Patton, W. 2002. Echo integration – trawl survey results for walleye pollock (*Theragra chalcogramma*) on the Bering Sea shelf and slope during summer 1999 and 2000. U.S. Department of Commerce, NOAA Technical Memorandum, NMFS-AFSC-125–126.
- Horne, J. K., and Walline, P. 2005. Spatial and temporal variance of walleye pollock (*Theragra chalcogramma*) in the eastern Bering Sea. *Canadian Journal of Fisheries and Aquatic Sciences*, 62: 1–10.
- ICES. 2005. Report of the Working Group on Fisheries Acoustics Science and Technology (WGFAST), 19–22 April 2005, Rome, Italy. ICES Document CM 2005/B: 05. 57 pp.
- MacLennan, D. N., Fernandes, P. G., and Dalen, J. 2002. A consistent approach to definitions and symbols in fisheries acoustics. *ICES Journal of Marine Science*, 59: 365–369.
- Misund, O. A. 1997. Underwater acoustics in marine fisheries and fisheries research. *Reviews in Fish Biology and Fisheries*, 7: 1–34.

- Mitson, R. B. 1995. Underwater noise of research vessels: review and recommendations. ICES Cooperative Research Report, 209. 61 pp.
- Mitson, R. B., and Knudsen, H. P. 2003. Causes and effects of underwater noise on fish-abundance estimation. *Aquatic Living Resources*, 16: 255–263.
- Ona, E., and Godø, O. R. 1990. Fish reaction to trawling noise: the significance for trawl sampling. *Rapports et Procès-verbaux des Réunions du Conseil International de l'Exploration de la Mer*, 189: 159–166.
- Ona, E., and Mitson, R. B. 1996. Acoustic sampling and signal processing near the seabed: the deadzone revisited. *ICES Journal of Marine Science*, 53: 677–690.
- Popper, A. N. 2003. Effect of anthropogenic sounds on fishes. *Fisheries*, 28: 24–31.
- Traynor, J. J. 1996. Target-strength measurements of walleye pollock (*Theragra chalcogramma*) and Pacific whiting (*Merluccius productus*). *ICES Journal of Marine Science*, 53: 253–258.
- Vabø, R., Olsen, K., and Huse, I. 2002. The effect of vessel avoidance of wintering Norwegian spring-spawning herring. *Fisheries Research*, 58: 59–77.
- Wardle, C. S. 1983. Fish reactions to towed fishing gears. *In* *Experimental Biology at Sea*, pp. 167–195. Ed. by A. MacDonald. Academic Press, New York. 414 pp.
- Wardle, C. S. 1993. Fish behaviour and fishing gear. *In* *Behaviour of Teleost Fishes*, pp. 609–643. Ed. by T. J. Pitcher. Chapman & Hall, New York. 533 pp.
- Zar, J. H. 1999. *Biostatistical Analysis*. Prentice-Hall, New Jersey. 663 pp.